# Safety data sheet



## 1. Identification

### Product identifier

# ULTRAMID® A3X2G7 BLACK 23187 POLYAMIDE

### Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Polymer
Recommended use: Polymer, for industrial processing only

### Details of the supplier of the safety data sheet

Company:

### Emergency telephone number

National emergency number:
+27 11 203 2420
International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

No need for classification according to GHS criteria for this product.

### Label elements

Globally Harmonized System (GHS)

The product does not require a hazard warning label in accordance with GHS criteria.

### Other hazards

According to UN GHS criteria

No specific dangers known, if the regulations/notes for storage and handling are considered.

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

Compound based on: polyamide (PA 66)

additives, fillers

flame retardant, phosphorus

Hazardous ingredients (GHS)
According to UN GHS criteria

phosphorus

| | |
|---|---|
| Content (W/W): >= 3 % - <= 6.5 % | Flam. Sol. 1 |
| CAS Number: 7723-14-0 | Aquatic Acute 3 |
| | Aquatic Chronic 3 |
| | H228, H402, H412 |

Zinc oxide

| | |
|---|---|
| Content (W/W): >= 0.1 % - <= 1 % | Aquatic Acute 1 |
| CAS Number: 1314-13-2 | Aquatic Chronic 1 |
| EC-Number: 215-222-5 | M-factor acute: 1 |
| INDEX-Number: 030-013-00-7 | M-factor chronic: 1 |
| | H400, H410 |

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

Avoid contact with the skin, eyes and clothing.

If inhaled:
If difficulties occur after dust has been inhaled, remove to fresh air and seek medical attention.

On skin contact:
Burns caused by molten material require hospital treatment.

On contact with eyes:
In case of contact with the eyes, rinse immediately for at least 15 minutes with plenty of water. If irritation develops, seek medical attention.

On ingestion:
Rinse mouth and then drink 200-300 ml of water. If difficulties occur: Seek medical attention.

### Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far

Hazards: No hazard is expected under intended use and appropriate handling.

### Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

### Extinguishing media

Suitable extinguishing media:
water spray, foam, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

### Special hazards arising from the substance or mixture

At temperatures of > 310 °C can be emitted: Ammonia, aqueous solution, Carbon monoxide, Carbon dioxide, Cyclopentanone, Hydrogen cyanide, Phosphine, amine derivatives, nitriles
Under special fire conditions traces of other toxic substances are possible. Formation of further decomposition and oxidation products depends upon the fire conditions.

### Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

High risk of slipping due to leakage/spillage of product.

### Personal precautions, protective equipment and emergency procedures

No special precautions necessary.

### Environmental precautions

No special precautions necessary.

### Methods and material for containment and cleaning up

For small amounts: Pick up with suitable appliance and dispose of.
For large amounts: Pick up with suitable appliance and dispose of.
For residues: Sweep/shovel up.
Dispose of absorbed material in accordance with regulations.

## 7. Handling and Storage

### Precautions for safe handling

Provide suitable exhaust ventilation at the drying process and in the area surrounding the melt outlet of processing machines. Closed containers should only be opened in well-ventilated areas. Ensure thorough ventilation of stores and work areas.

Any short stoppages in production, it is recommended that you inject material into the mould not purge an air shot. Any molten material drooling from the machine nozzle or hot runner nozzles can self-ignite when in open atmosphere.
It is therefore advisable to dispose of purgings etc into water containers.

For additional guidelines and recommendations see the product-specific "Processing Data Sheet".

Protection against fire and explosion:
Take precautionary measures against static discharges.

### Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds.

Suitable materials for containers: Low density polyethylene (LDPE), High density polyethylene (HDPE), Aluminium, Carbon steel (Iron)

Storage stability:
Protect against moisture.

### Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

7803-51-2: Phosphine
TWA value 0.1 ppm
STEL value 0.3 ppm

Given suitable ventilation, it can be assumed that the threshold limits will not be reached.

### Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if breathable aerosols/dust are formed. Wear respiratory protection if ventilation is inadequate. (Particle filter EN 143 P2 or FFP2)

Hand protection:
Use additional heat protection gloves when handling hot molten masses (EN 407), e.g. of textile or leather.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Do not inhale gases/vapours/aerosols. When using, do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. After use of gloves apply skin-cleaning agents and skin cosmetics.

## 9. Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| State of matter: | solid | |
| Form: | granules | |
| Colour: | black | |
| Odour: | odourless | |
| Odour threshold: | not applicable | |
| melting range: | 260 °C<br>(1,013 hPa) | (DIN 53765) |

| | | |
|---|---|---|
| Boiling range: | The substance / product decomposes therefore not determined. | |
| Sublimation point: | No applicable information available. | |
| Flammability: | not self-igniting | (derived from flash point) |
| Flammability of Aerosol Products: | not applicable, the product does not form flammable aerosoles | |
| Lower explosion limit: | For solids not relevant for classification and labelling. | |
| Upper explosion limit: | For solids not relevant for classification and labelling. | |
| Flash point: | not applicable | |
| Auto-ignition temperature: | > 350 °C | (ASTM D1929) |
| Thermal decomposition: | > 310 °C (TGA)<br>To avoid thermal decomposition, do not overheat. | |
| pH value: | not applicable | |
| Viscosity, kinematic: | not applicable, the product is a solid | |
| Viscosity, dynamic: | not applicable, the product is a solid | |
| Solubility in water: | insoluble<br>(20 °C, 1,013 hPa) | (OECD Guideline 105) |
| Partitioning coefficient n-octanol/water (log Kow): | not applicable | |
| Vapour pressure: | not applicable | |
| Relative density: | Study does not need to be conducted. | |
| Density: | 1.30 - 1.60 g/cm3<br>(20 °C, 1,013 hPa) | (EN ISO 1183-1) |
| Relative vapour density (air): | not applicable | |
| Particle characteristics | | |
| Particle size distribution: | spheroidal - | |
| Specific Surface Area: | 0.0 m²/g | (MSSA, ISO 9227) |

## 9.2. Other information

**Information with regard to physical hazard classes**

| | | |
|---|---|---|
| Explosives | | |
| Explosion hazard: | not explosive | |
| Oxidizing properties | | |
| Fire promoting properties: | not fire-propagating | |
| Pyrophoric properties | | |
| Self-ignition temperature: | | Test type: Spontaneous self-ignition at room-temperature. |
| | not self-igniting | |
| Self-heating substances and mixtures | | |

| | | |
|---|---|---|
| Self heating ability: | It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2. | (VDI 2263, sheet 1, 1.4.1 (May 1990)) |
| Corrosion to metals | | |
| | No corrosive effect on metal. | |

**Other safety characteristics**

| | | |
|---|---|---|
| Bulk density: | 500 - 800 kg/m3<br>(20 °C, 1,013 hPa) | (DIN 53466) |
| Evaporation rate: | | |
| | The product is a non-volatile solid. | |

## 10. Stability and Reactivity

### Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

### Chemical stability

The product is stable if stored and handled as prescribed/indicated.
The product is chemically stable.

### Possibility of hazardous reactions

No hazardous reactions known.
The product is chemically stable.

### Conditions to avoid

Temperature: > 310 °C
See SDS section 7 - Handling and storage.

### Incompatible materials

Substances to avoid:
No substances known that should be avoided.

### Hazardous decomposition products

Hazardous decomposition products:
Ammonia, aqueous solution, Carbon monoxide, Carbon dioxide, Cyclopentanone, Hydrogen cyanide, Phosphine
amines, nitriles

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Contact with molten product may cause thermal burns.

Irritation

Assessment of irritating effects:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Serious eye damage/irritation: May cause mechanical irritation.

Respiratory/Skin sensitization

Assessment of sensitization:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Carcinogenicity

Assessment of carcinogenicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Developmental toxicity

Assessment of teratogenicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
not applicable

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

## 12. Ecological Information

### Toxicity

Assessment of aquatic toxicity:
The product has not been tested. The statement has been derived from the structure of the product. There is a high probability that the product is not acutely harmful to aquatic organisms.

### Persistence and degradability

Assessment biodegradation and elimination (H2O):
Experience shows this product to be inert and non-degradable.
The product is virtually insoluble in water and can thus be separated from water mechanically in suitable effluent treatment plants.

Elimination information:
Poorly biodegradable.

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.

### Bioaccumulative potential

Assessment bioaccumulation potential:
Does not significantly accumulate in organisms.

Bioaccumulation potential:
The product will not be readily bioavailable due to its consistency and insolubility in water.

### Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: Adsorption to solid soil phase is not expected.

### Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

### Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

### Additional information

Adsorbable organically-bound halogen (AOX):
This product contains no organically-bound halogen.

Other ecotoxicological advice:
The product is a polymeric compound.

## 13. Disposal Considerations

### Waste treatment methods

Check for possible recycling.
Incinerate in suitable incineration plant, observing local authority regulations.

Contaminated packaging:
Packs must be completely emptied.
Completely emptied packagings can be given for recycling.

## 14. Transport Information

**Land transport**

ADR

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

RID

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |

| | |
|---|---|
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

## Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Regulation on Registration, Evaluation, Authorisation and Restriction of Chemicals (EU)
Restrictions of Regulation (EC) No 1907/2006, Annex XVII, do not apply for the intended use(s) of the product given in this SDS.

## 16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Flam. Sol. | Flammable solids |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| H228 | Flammable solid. |
| H402 | Harmful to aquatic life. |
| H412 | Harmful to aquatic life with long lasting effects. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.